# THE FINITE PART OF A SEPARATED LOCALLY FINITE TYPE MORPHISM IS AN ALGEBRAIC SPACE, ETALE OVER Y

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*Editorial note.* No omission markers in-span.

*Allowed standard prerequisites (editorial).*

- Sheaf condition, diagonal, cover. lemma-finite-sheaf (carded L26650), lemma-finite-diagonal (carded L26653), lemma-finite-surjective-etale-cover (carded L26655). The three algebraic-space axioms, each verified by a carded lemma.
- Restricting to the quasi-finite locus. lemma-finite-lives-on-locally-quasi-finite-part (carded L26651). Replaces X by its locally quasi-finite open.
- Etale-local detection. Properties of Spaces, lemma etale-local (cross-chapter, named). Etaleness of the south-west arrow from the triangle.

*Source reference.* Original lemma 04PK: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1744–1744. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 04PG: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1647–1647. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 04PE: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1550–1550. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 04QG: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1848–1848. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 025Y: focused-seeds/geometry/S9-smg-finite-algebraic-space/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 03FS: focused-seeds/geometry/S9-smg-finite-algebraic-space/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Proposition 0.1.** *Let $S$ be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over $S$ which is separated and locally of finite type. Then $(X/Y)_{fin}$ is an algebraic space. Moreover, the morphism $(X/Y)_{fin} \to Y$ is étale.*

*Proof.* By Lemma 04PG we may replace $X$ by the open subscheme which is locally quasi-finite over $Y$. Hence we may assume that $f$ is separated and locally quasi-finite. We will check the three conditions of Spaces, Definition 025Y. Condition (1) follows from Lemma 04PE. Condition (2) follows from Lemma 04PK. Finally, condition (3) follows from Lemma 04QG. Thus $(X/Y)_{fin}$ is an algebraic space. Moreover, that lemma shows that there exists a commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & (X/Y)_{fin} \\ & \searrow \quad \swarrow & \\ & Y & \end{array}$$

with horizontal arrow surjective and étale and south-east arrow étale. By Properties of Spaces, Lemma 03FS this implies that the south-west arrow is étale as well. □